

# Fathima Nafra

BSc (Hons) in Computer Science (Specialization: Data Science) - Undergraduate

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## Profile Summary

Computer Science undergraduate at the University of Kelaniya, Faculty of Computing & Technology with a strong interest in data analysis, data engineering, and machine learning. Experienced in collecting, processing, and analyzing data to support real-world decision-making. Skilled in working with datasets, building data workflows, and applying machine learning techniques to uncover patterns and insights. Motivated to continuously develop technical and analytical skills, with a goal of contributing to practical, data-driven solutions.

## Education

BSc. (Hons) in Computer Science (Specialization: Data Science) 2023 - Present  
Faculty of Computing & Technology, University of Kelaniya  
Current GPA: 3.62/4.00

Diploma in ICT & English 2022-2023  
ICBT Galle Campus

G.C.E. Advanced Level - Physical Science Stream Completed: 2021  
Results: Physics - B, Chemistry - B, Combined Mathematics - A  
Rippon Girls' College, Galle

## Projects

- 1. Loan Default Risk Approval Prediction System** 2026
  - Developed LightGBM classification & regression models to predict loan approval and generate risk scores.
  - Improved model performance through preprocessing, feature engineering, and hyperparameter tuning.
  - Applied SHAP for model interpretability and feature importance analysis.
  - Built a Flask web app for real-time prediction and user interaction.
  - *Technologies:* Python, LightGBM, SHAP, scikit-learn, Flask
- 2. End-to-End Azure Data Engineering Pipeline** 2026
  - Designed and implemented a scalable data pipeline with automated ingestion into Azure SQL.
  - Structured data processing using Medallion architecture (Bronze - Silver layers).
  - Performed efficient data transformation and processing using PySpark in Databricks.
  - *Technologies:* Azure Data Factory, Azure SQL, Databricks, PySpark
- 3. Data Warehouse and Analytics Project** 2026
  - Built a comprehensive data warehouse using Medallion Architecture to process, transform, and integrate large multi-source datasets.
  - *Technologies:* SQL, SQL Server Management Studio

#### 4. AI-Based Crypto & Forex Signal Generator

2026

- Built a full-stack app generating real-time trading signals using RSI, EMA, MACD, and ML models.
- Developed backend APIs with Flask and frontend UI using React (Vite).
- *Technologies:* Python, Flask, React, Pandas, Scikit-learn

Collaborated in a team of 2 to design and develop the system, contributing to both frontend and backend development.

#### 5. HallEase - Event & Hall Management System

2<sup>nd</sup> year - Group Project

- Developed a web-based system for hall booking, scheduling, and admin management.
- Improved booking efficiency and reduced scheduling conflicts.
- *Technologies:* TypeScript, Next.js, Tailwind CSS, Supabase

#### 6. Netflix Movies & TV Shows Data Analysis

2026

- Analyzed Netflix dataset using SQL to extract insights on trends, genres, and distribution.
- Built Power BI dashboard to visualize KPIs and content patterns.
- *Technologies:* SQL, Power BI

### Certifications

- Azure Fundamentals Course for Beginners - ScholarHat
- Machine Learning with Python Free Course - ScholarHat
- AWS AI Practitioner Challenge - Udacity
- Power BI for Beginners - Simplilearn
- SQL Server Foundations Course - ScholarHat
- Python 101 for Data Science - Cognitive Class
- AWS Certified Machine Learning Engineer - Associate - Udemy (Reading)

### Technical Skills

- Programming Languages: Python, R, JavaScript
- Web Development & Frameworks: React, Flask, HTML, CSS
- Machine Learning & Data Analysis: scikit-learn, pandas, NumPy, Matplotlib, Seaborn, Streamlit
- Data Engineering & Big Data: ETL pipelines, Azure Data Factory, Azure Databricks, PySpark
- Databases: MySQL, Azure SQL, PostgreSQL, Supabase, PL/SQL
- Cloud & Development Tools: Git, GitHub, Azure, AWS (basic), Jupyter Notebook, Power BI

### Soft Skills

Analytical Thinking | Problem Solving | Clear Communication | Adaptability | Cross-Functional Collaboration